

Article

A Low-Power Low-IF BLE Receiver Front-End with a Common-Gate TIA and Gm-C Complex Filter for Body Area Network Applications

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Abstract

In this article, a low-power low-intermediate-frequency (Low-IF) receiver front-end is presented for Bluetooth Low Energy (BLE) body area network (BAN) applications. The receiver employs an input matching network, an inductorless self-biased inverter-based low-noise transconductance amplifier (LNTA), a single-balanced passive mixer, a common-gate transimpedance amplifier (TIA), and a Gm-C complex filter for image suppression. Native MOS devices are adopted to support low-voltage operation and reduce static power consumption. The interstage on-chip coupling capacitor between the RF front-end and the TIA is removed by aligning the DC operating points of the two stages. The receiver front-end is implemented in a 55 nm standard CMOS process and occupies an active area of 0.081 mm², excluding bonding pads. Post-layout simulations show that the proposed design achieves 45.2 dB gain, 7.2 dB noise figure, and 28.1 dB image rejection ratio over the 2.4–2.48 GHz band, while consuming 537 μ W. The proposed front-end is suitable for low-power BLE BAN sensor nodes.

Keywords: Bluetooth Low Energy; body area network; receiver front-end; low-IF receiver; transimpedance amplifier; complex filter; low-power CMOS

1. Introduction

With the rapid development of 5G technology and the widespread deployment of Internet of Things (IoT) applications, wireless sensor nodes are being used on an increasingly large scale [1,2]. Under this trend, energy supply, battery lifetime, and system cost have become key constraints in practical IoT devices. Therefore, the design of low-power, low-cost, and long-lifetime wireless communication systems has become an important research topic [3].

A body area network (BAN) is a short-range wireless communication network built around the human body, with a typical communication distance of 0.01–3 m. BAN nodes are usually distributed on or inside the human body and are used for continuous monitoring and transmission of physiological and behavioral information. As shown in Figure 1, smart healthcare BAN systems may include sensor nodes for electroencephalogram (EEG), electrocardiogram (ECG), electromyogram (EMG), blood pressure (BP), oxygen saturation (SpO₂), and inertial sensing, which communicate with smartphones or other portable terminals through Bluetooth Low Energy (BLE). A typical BAN terminal node consists of sensors,



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an analog-to-digital converter (ADC), a microcontroller unit (MCU), memory, and a radio-frequency (RF) transceiver, and may also operate together with energy-harvesting and power-management circuits. Although BAN communication distances are short, the energy budget is highly constrained because the nodes are usually powered by miniature batteries or other energy-limited sources [4,5]. In practice, a BAN node remains in standby mode for most of the time (97%), while wireless data transmission dominates the power consumption during active operation (3%) [6]. Therefore, reducing the power consumption of the RF transceiver, especially that of the receiver front-end, is essential for extending node lifetime.

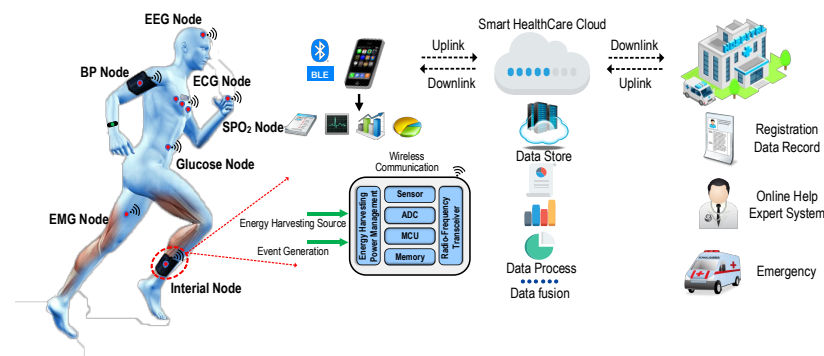


Figure 1. Smart healthcare monitoring system for body area networks.

BLE is a representative short-range wireless standard and has been widely adopted in IoT sensor devices. BLE receiver specifications are relatively relaxed compared with many other wireless standards. For 1 MSym/s uncoded data, the sensitivity requirement is -70 dBm, and the corresponding NF, IIP3, and image-rejection constraints are summarized in Table 1. This allows low-power BLE receivers to adopt comparatively simple architectures [7,8]. Nevertheless, the receiver front-end remains a critical and power-sensitive block, because insufficient gain, excessive NF, or inadequate image rejection directly degrades receiver performance in practical 2.4 GHz environments.

Many 2.4 GHz receiver architectures have been reported in recent years [7,9–11,11–15]. Zero-IF receivers provide high integration and simple frequency planning, but they usually require additional circuits to mitigate I/Q mismatch, DC offset, and flicker-noise-related issues. By contrast, the low-intermediate-frequency (Low-IF) architecture can alleviate many of these drawbacks, although it introduces the challenge of image rejection. In addition, Low-IF and heterodyne receivers support LO phase manipulation and can be extended to phase-shifting and beamsteering architectures. Recent studies on real-time beamsteering based on direct digital synthesizer–phase-locked loop (DDS-PLL) architectures further illustrate this flexibility [16–19]. For BLE body area network applications, however, low-power techniques such as low-voltage operation and compact circuit implementation remain the primary design concerns [20–22].

Based on the above considerations, this work presents a low-power and compact Low-IF BLE receiver front-end for body area network (BAN) applications. The contribution of this work does not lie in the adoption of the Low-IF architecture itself, but rather in the BAN-oriented circuit co-design under stringent power and area constraints. First, an inductorless self-biased inverter-based LNTA employing native MOS devices is introduced to enable low-voltage operation and reduce static power consumption. Second, by properly aligning the DC operating points of the RF front-end and the subsequent TIA, the conventional interstage on-chip coupling capacitor is eliminated, thereby reducing both silicon area and parasitic loading. Third, a common-gate current-mode TIA and a Gm-C complex filter are co-designed to realize current-to-voltage conversion, gain distribution, low-pass shaping, and image rejection within a compact Low-IF signal chain. In this way,

the proposed front-end achieves a balanced tradeoff among gain, noise figure, image rejection ratio, and power consumption, which makes it suitable for low-power BLE BAN sensor nodes.

The remainder of this article is organized as follows. Section 2 presents the system requirements and receiver architecture selection. Section 3 describes the circuit design of the RF front-end, TIA, and complex filter. Section 4 presents the simulation results. Section 5 discusses the design implications, advantages, and limitations of the proposed front-end. Finally, Section 6 concludes this article.

Table 1. BLE receiver specifications for 1-MSym/s uncoded data [7,8].

Parameter	Specifications
Operating frequency	2.4–2.4835 GHz
Channel BW	1 MHz
Channel spacing	2 MHz
Sensitivity with 1 MSym/s uncoded data	<−70 dBm
Interference performance	
Co-channel interference, $C/I_{\text{co-channel}}$	21 dB
Adjacent (1 MHz) interference, $C/I_{1\text{MHz}}$	15 dB
Adjacent (2 MHz) interference, $C/I_{2\text{MHz}}$	−17 dB
Adjacent (>3 MHz) interference, $C/I_{3\text{MHz}}$	−27 dB
Image frequency interference, C/I_{Image}	−9 dB
Noise figure	<30 dB
IIP3	>−31 dB
Image rejection ratio	>21 dB

2. System Requirements and Receiver Architecture

2.1. BAN-Oriented Design Targets

For body area network (BAN) applications, the BLE receiver front-end must support short-range reliable communication under a strict power budget. In practical BAN scenarios, body shadowing, body coupling, and fading may introduce additional attenuation and signal fluctuation; thus, sufficient robustness is required under low-power constraints. According to the BLE specification, the transmitter output power typically ranges from −20 dBm to 0 dBm. Based on the Friis transmission equation, the received power can be expressed as

$$P_{RX} = P_{TX} + A_{TX} + A_{RX} - 10 \log_{10} \left(\frac{4\pi D}{\lambda} \right)^2 \quad (1)$$

where P_{TX} is the transmitter output power, A_{TX} and A_{RX} are the transmitter and receiver antenna gains, respectively, D is the communication distance, and λ is the wavelength. In this work, both antenna gains are assumed to be 0 dBi and the operating frequency is around 2.4 GHz. As shown in Figure 2, when the transmitter power is −20 dBm, the received signal is approximately −70 dBm at a distance of about 3 m, which is sufficient for typical BAN applications. Therefore, the receiver sensitivity target is set close to −70 dBm, aiming at a balanced tradeoff among sensitivity, gain, noise, linearity, and power.

Based on the BLE physical-layer requirements, the main receiver-level specifications are summarized in Table 1. Combining these requirements with the target BAN communication range, the adopted front-end design specifications are listed in Table 2. The operating

band is 2.4–2.4835 GHz, and a 1 MHz Low-IF architecture is adopted. The front-end is designed to achieve a sensitivity better than -70 dBm, NF below 15 dB, IIP3 above -28 dBm at the maximum-gain setting minus 6 dB, IRR above 24 dB, and total power consumption below 1 mW. These targets serve as the practical design constraints for the subsequent RF front-end, TIA, and complex filter design.

Table 2. Receiver front-end design specifications adopted in this work.

Parameter	Specification
Operating frequency	2.4–2.4835 GHz
Architecture	1 MHz Low-IF
Sensitivity	< -70 dBm
Noise figure (NF)	< 15 dB
IIP3 at maximum gain -6 dB	> -28 dBm
Image rejection ratio (IRR)	> 24 dB
Receiver front-end power	< 1 mW

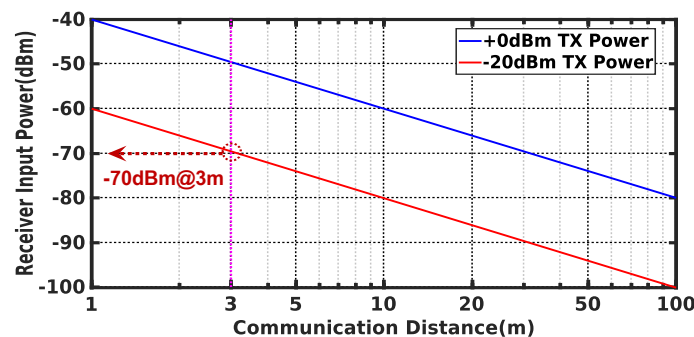


Figure 2. Estimated relationship between communication distance and receiver input power for two transmitter power levels.

2.2. Architecture Selection and Proposed Receiver Front-End

Several receiver architectures can be considered for low-power BLE applications, including superheterodyne, zero-IF, and Low-IF structures. Superheterodyne receivers provide strong selectivity but are less suitable for highly integrated BAN nodes because of their complexity and power overhead. Zero-IF receivers are compact, but they are sensitive to DC offset, LO leakage, and flicker noise. Therefore, a Low-IF architecture is adopted in this work as a compromise between implementation simplicity and robustness, although it requires image rejection. For the target 1 MHz intermediate frequency, the desired BLE signal can be shifted away from DC while enabling compact analog baseband implementation [20,21].

Recent BLE standards also support localization techniques such as angle-of-arrival (AoA) estimation. In such applications, hardware impairments may degrade the estimation accuracy and often require compensation. However, this work focuses on low-power BLE receiver front-end design for BAN applications rather than localization-oriented calibration methods [23–25].

Figure 3 shows the proposed BLE receiver front-end, which consists of an input matching network, an RF front-end, a common-gate current-mode TIA, and a Gm-C complex filter. The matching network, self-biased LNTA, and passive mixer form a low-power RF chain that down-converts the 2.4–2.4835 GHz BLE signal to a 1 MHz low-IF current signal. The TIA performs current-to-voltage conversion and preliminary low-pass shaping, while the Gm-C complex filter provides image suppression and improved selectivity. By adopting a DC-compatible interface between the RF front-end and the TIA, the conventional interstage coupling capacitor is removed, reducing both area and parasitic

loading. Therefore, the proposed architecture provides a compact and low-power solution for BAN-oriented BLE receiver front-ends.

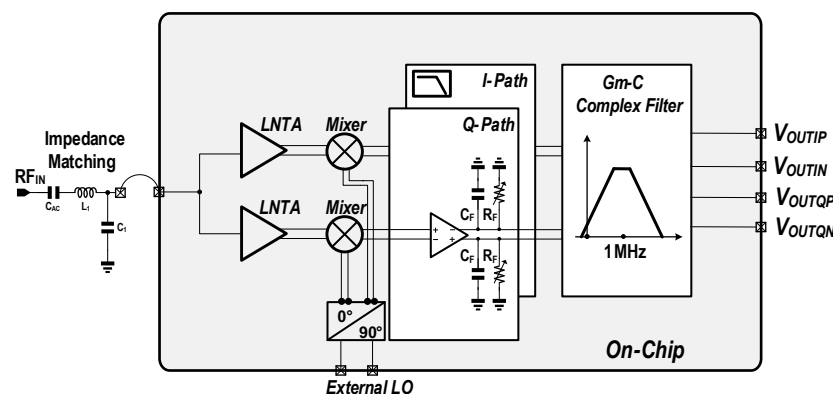


Figure 3. Architecture of the proposed BLE receiver front-end.

3. Detailed Circuit Design of the Proposed BLE Receiver Front-End

3.1. RF Front-End

Figure 4 shows the proposed receiver front-end, which consists of an input matching network, a self-biased LNTA, and a single-balanced passive mixer. The front-end is designed as a compact and low-power RF signal chain for BAN-oriented BLE applications. The matching network, formed by C_{AC} , L_1 , and C_1 , provides input matching to the $50\ \Omega$ source impedance and contributes passive voltage gain. The LNTA adopts a self-biased inverter-based topology, where a native NMOS device is used to reduce the operating voltage and power consumption. By aligning the DC operating points of the RF front-end and the following TIA, the conventional interstage coupling capacitor is removed, leading to reduced area and parasitic loading. The single-balanced passive mixer performs current-mode down-conversion using 50%-duty-cycle non-overlapping LO signals, offering lower LO-related power than a double-balanced structure at the expense of some gain and noise degradation. In this way, the proposed RF front-end integrates matching, amplification, and down-conversion in a compact implementation with reduced power and area. The down-converted low-IF current is then delivered to the following TIA and image-rejection stage. The large-size PMOS device MP1 is implemented with a multi-finger layout to alleviate parasitic effects. The influence of layout parasitics is included in the post-layout simulations, while process variations are partially reflected by corner simulations; the 60 nm channel length is chosen to maintain high transconductance under low-voltage operation. This self-biased scheme is adopted mainly to reduce circuit complexity and power consumption, and its robustness is evaluated through post-layout and process-corner simulations. Based on this topology, the gain, noise, linearity, and power tradeoffs of the proposed front-end are analyzed in the following subsection.

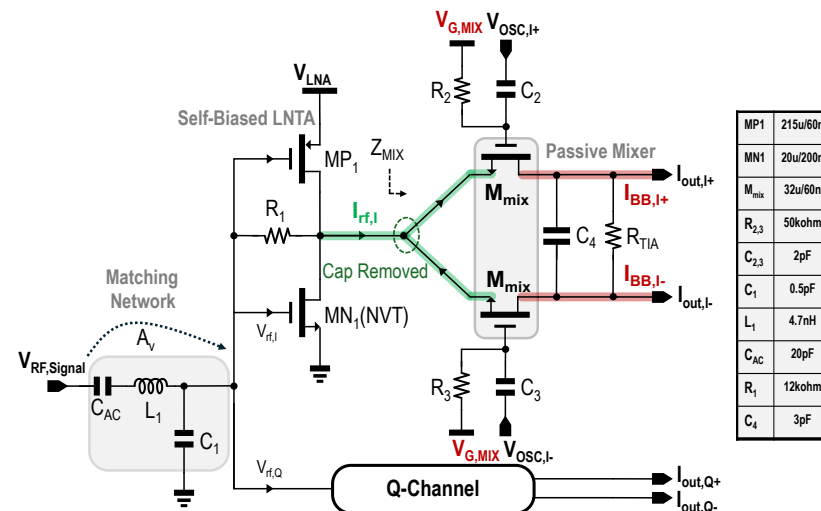


Figure 4. Schematic of the proposed RF front-end consisting of an input matching network, a self-biased inverter-based LNTA, and a single-balanced passive mixer.

3.1.1. Impedance Matching

As shown in Figure 5, the input matching network can be represented by the equivalent circuits in Figure 5a,b, where C_{in} denotes the equivalent input capacitance composed of C_1 and the LNTA gate capacitance, and V_S and R_S represent the antenna source and source impedance, respectively. For narrowband analysis, the network can be simplified to the series equivalent circuit in Figure 5b, with equivalent capacitance C_{ser} and resistance R_{ser} .

Under the maximum power transfer condition,

$$\frac{V_{out}^2}{2R_{out}} = \frac{V_S^2}{8R_S}, \quad V_{out} = A_v V_{in}, \quad V_{in} = \frac{V_S}{2} \quad (2)$$

and the passive voltage gain can be written as

$$A_v = \sqrt{\frac{R_{out}}{R_S}} \quad (3)$$

When L_1 resonates with C_{ser} , the resonant angular frequency and quality factor are

$$\omega_0 = \frac{1}{\sqrt{L_1 C_{ser}}}, \quad Q_C = \frac{\sqrt{L_1/C_{ser}}}{R_{ser}} \quad (4)$$

Accordingly, the passive voltage gain is

$$|A_v| = \left| \frac{V_{out}}{V_{in}} \right| = \sqrt{1 + \frac{Q_C^2}{4}} \quad (5)$$

which indicates that the matching network provides impedance transformation together with passive voltage boosting.

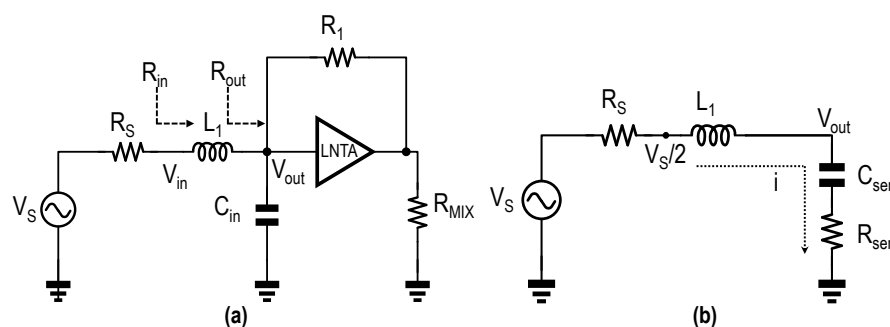


Figure 5. Input impedance matching of the receiver front-end: (a) L_1 and C_{in} act as the impedance transformation network; (b) narrowband equivalent circuit.

3.1.2. Gain

Figure 6 shows the small-signal equivalent circuit of the I-channel of the proposed receiver front-end, including the following TIA stage. In this model, the LNTA is represented by the current source $g_m V_{in}$ and the output resistance R_{out} . The equivalent impedance looking into the mixer is denoted by Z_{MIX} , while R_{sw} represents the on-resistance of the passive mixer switches and R_{sh} is the equivalent shunt resistance associated with harmonic translation in the mixer. The input impedance of the common-gate TIA is denoted by R_{TIA} , and C_D represents the harmonic-bypass capacitance at the mixer output, including C_4 and the input capacitance of the TIA. In addition, I_{BBp} and I_{BBn} denote the baseband amplification currents in the TIA, whereas R_{BB} and C_{BB} form the output low-pass network of the TIA. The node voltages V_x , $V_{yp/n}$, and $V_{outp/n}$ correspond to the LNTA output, mixer output, and TIA output voltages, respectively.

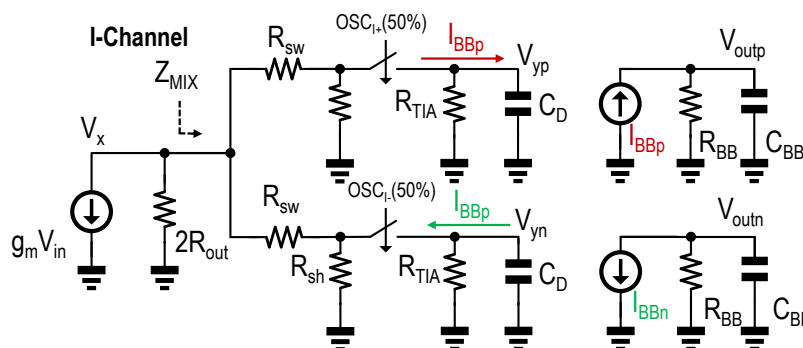


Figure 6. Small-signal equivalent circuit of the I-channel of the receiver front-end, including the TIA Stage.

Based on the equivalent models of passive mixers and N-path circuits reported in [26–30], the equivalent mixer input impedance can be approximated as

$$Z_{MIX}|_{(\omega_{LO}+\omega_{IF})} \approx R_{sw} + \left(\frac{2Z_{BB}}{\pi^2} \parallel R_{sh} \right) \quad (6)$$

where

$$Z_{BB} = \frac{1}{sC_D} \parallel R_{TIA}, \quad R_{sh} \approx \frac{2}{3}(2R_{out} + R_{sw}) \quad (7)$$

Accordingly, the voltage gain at the LNTA output node can be written as

$$A_{Vx}|_{(\omega_{LO}+\omega_{IF})} \approx g_m(2R_{out} \parallel Z_{MIX}) \quad (8)$$

The equivalent down-conversion transconductance of the receiver front-end can be approximated by

$$G_m = \frac{I_{BB}}{V_{in}} \Big|_{DC} = \frac{I_{BBp} - I_{BBn}}{V_{in}} \approx g_m \frac{R_{out}}{R_{TIA} + 2(R_{out} + R_{sw})} \frac{4}{\pi} \quad (9)$$

Based on (9), the voltage gains at the mixer output and TIA output can be expressed as

$$A_{Vy}|_{DC} = A_{Vyp} - A_{Vyn} \approx G_m R_{TIA} \quad (10)$$

and

$$A_{Vout}|_{DC} = A_{Voutp} - A_{Voutn} \approx G_m R_{BB} \quad (11)$$

Equations (6)–(11) show that the overall gain of the proposed receiver front-end is mainly determined by the LNTA transconductance and the TIA-related output impedance. Increasing the mixer switch size can reduce R_{sw} and improve gain, but it also increases LO-driving power. Therefore, the receiver gain should be optimized together with power consumption and noise performance to achieve a balanced front-end design.

3.1.3. Noise

Figure 7 shows the equivalent noise model of the proposed receiver front-end. The dominant noise sources considered here are the source-resistance thermal noise, the LNTA channel-current noise, the mixer switch thermal noise, and the equivalent input-referred noise of the following TIA. These noise sources can be expressed as

$$\overline{V_{n,Rs}^2} = 4kTR_S, \quad \overline{I_{n,gm}^2} = 4kT\gamma_1 g_m, \quad \overline{V_{n,sw}^2} = 4kTR_{sw}, \quad \overline{V_{n,TIA}^2} \approx 4kT\gamma_2 R_{TIA} \quad (12)$$

where k is the Boltzmann constant, T is the absolute temperature, and γ_1 and γ_2 are the MOS thermal-noise coefficients. The thermal noise of the LNTA self-bias resistor is neglected because its contribution is largely canceled by the balanced I/Q structure.

Based on the gain relations in (2)–(11), the receiver front-end noise factor can be approximated as

$$F = 1 + \frac{\gamma_1}{R_S A_v^2 G_m} + \frac{R_{sw}}{R_S A_v^2 G_m^2 R^2} + \frac{(R + R_{sw})^2}{R_S A_v^2 G_m^2 R^2 \beta \gamma_2 R_{TIA}} + \frac{a \gamma_1}{R_S A_v^2 G_m} + a + \frac{a R_{sw}}{R_S A_v^2 G_m^2 R^2} \quad (13)$$

with

$$\beta = \frac{2}{\pi^2}, \quad a = \left(\frac{\pi^2}{4} - 1 \right), \quad G_m = g_m, \quad R = R_{out} \quad (14)$$

Equation (13) shows that increasing the passive matching gain, the LNTA transconductance, or the TIA gain can reduce the noise contribution of the following stages. However, larger gain generally degrades linearity, indicating a tradeoff among gain, noise, and linearity in the proposed receiver front-end.

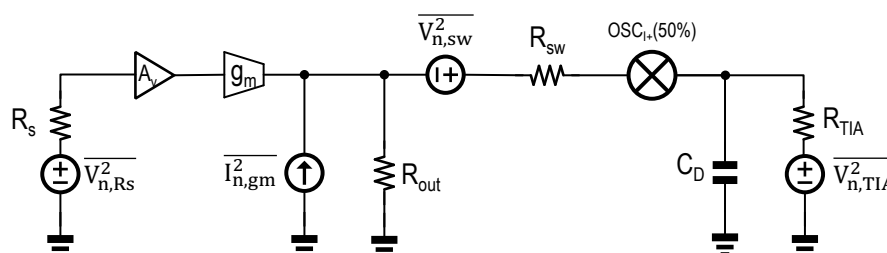


Figure 7. Equivalent linear time-invariant noise model of the receiver front-end.

3.1.4. Linearity

The linearity of the receiver front-end can be characterized by the input third-order intercept point (IIP₃). IIP₃ can be derived from the third-order intermodulation term. In this work, the front-end linearity is assumed to be mainly limited by the first LNTA stage. The nonlinear output current of the LNTA consists of two contributions: one directly related to the input signal v_{in} and the other associated with the cubic nonlinearity of the LNTA output node v_x in Figure 6. Accordingly, the LNTA drain current can be expressed as

$$i_{ds} = \alpha_1 v_{in} + \alpha_2 v_{in}^3 + \alpha_3 v_x^3 \quad (15)$$

where α_1 , α_2 , and α_3 denote the linear transconductance, third-order transconductance, and third-order output conductance of the LNTA, respectively, i.e.,

$$\alpha_1 = g_m, \quad \alpha_2 = g_{m3}, \quad \alpha_3 = g_{o3} \quad (16)$$

Assuming that the input signal with amplitude A is converted by the LNTA into a current signal and then transformed into an output voltage by the TIA, the third-order intermodulation components at the receiver output can be written as

$$v_{o3,\alpha_2} = \frac{3}{4} g_{m3} A^3 I_{BB} R_{BB}, \quad v_{o3,\alpha_3} = \frac{3}{4} g_{o3} A_{Vx}^3 A^3 I_{BB} R_{BB} \quad (17)$$

while the fundamental output component is

$$v_{o1} = A g_m I_{BB} R_{BB} \quad (18)$$

Therefore, the total third-order distortion term is

$$v_{o3} = v_{o3,\alpha_2} + v_{o3,\alpha_3} = \frac{3}{4} g_{m3} A^3 I_{BB} R_{BB} + \frac{3}{4} g_{o3} A_{Vx}^3 A^3 I_{BB} R_{BB} \quad (19)$$

According to the definition of IIP₃, the input amplitude at which the extrapolated fundamental component equals the third-order intermodulation component yields

$$\text{IIP}_3 = \sqrt{\frac{4g_m}{3(g_{m3} + g_{o3}A_{Vx}^3)}} \quad (20)$$

Equation (20) indicates that increasing the LNTA transconductance can improve the front-end linearity, but this requires higher current consumption. In addition, an excessively large front-end gain A_{Vx} aggravates the cubic nonlinearity contribution and reduces IIP₃. Therefore, the proposed receiver front-end exhibits a fundamental tradeoff among gain, linearity, and power consumption, which should be jointly optimized in the circuit design.

3.1.5. Pre-Layout Simulation Results of the Receiver Front-End

The proposed receiver front-end, including the LNTA and passive mixer, was pre-layout simulated in Cadence Virtuoso using SpectreRF based on the HLMC 55 nm 1P4M CMOS process under the TT corner at 27 °C. The simulated input matching, down-conversion gain, linearity, and noise figure are summarized in Figure 8. With a 0.6 V LNTA supply, an LO frequency of 2.45 GHz, and an LO amplitude of 300 mV, the front-end achieves $S_{11} < -10$ dB across the BLE band, with a minimum value of -21.54 dB at 2.4 GHz. Under an RF input frequency of 2.451 GHz and an input power of -60 dBm, the proposed design provides a down-conversion voltage gain of 23.9 dB and a noise figure of 2.67 dB at an IF of 1 MHz. Two-tone harmonic-balance analysis further gives an input third-order intercept point of -7.68 dBm. Overall, the pre-layout results demonstrate that the proposed

front-end achieves satisfactory input matching together with favorable gain and noise performance, while maintaining acceptable linearity.

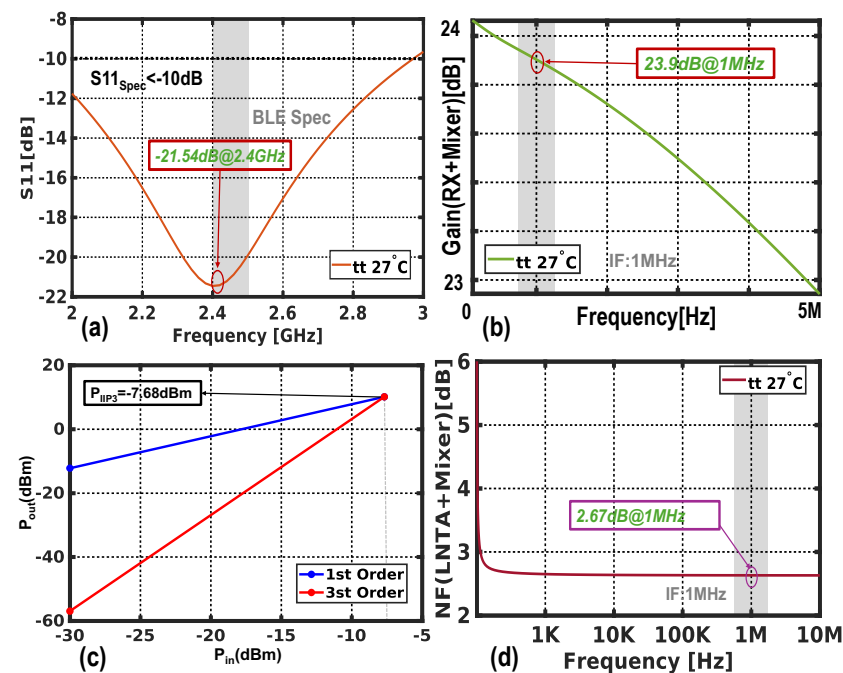


Figure 8. Pre-layout simulated performance of the proposed receiver front-end: (a) input reflection coefficient S_{11} , (b) down-conversion voltage gain, (c) input third-order intercept point (IIP_3), and (d) noise figure (NF).

3.2. Common-Gate TIA

As shown in Figure 9, a low-gain common-gate current-mode TIA is employed to interface the current-mode passive mixer with the following image-rejection filter. The fully differential I/Q structure provides current-to-voltage conversion, gain tuning, and preliminary low-pass filtering. Owing to the inherently low input impedance of the common-gate topology, the proposed TIA can be directly cascaded with the passive mixer.

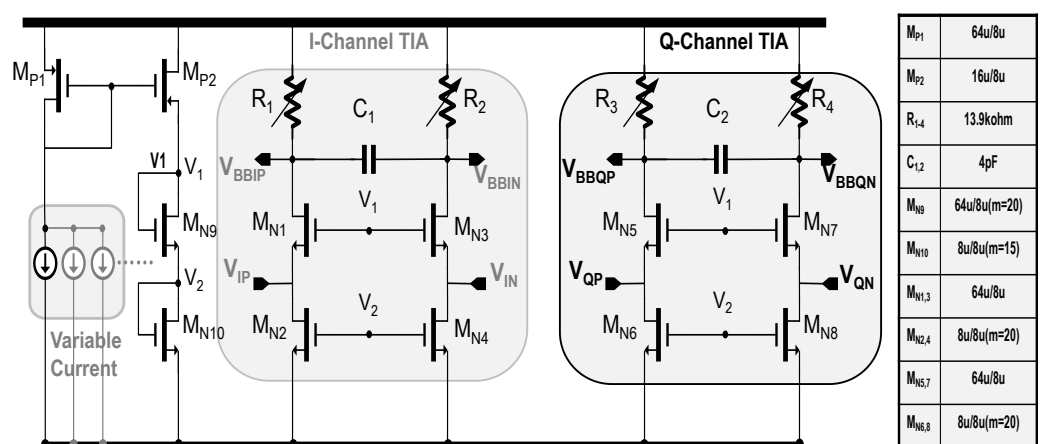


Figure 9. Common-gate transimpedance amplifier used for current-to-voltage conversion and variable-gain amplification in the proposed Low-IF signal chain.

For midband analysis, the equivalent output resistance is approximated by

$$R_{eq} \approx R_L \quad (21)$$

and the input impedance can be written as

$$Z_{in} \approx \frac{1}{g_{m1} + g_{mb1}} \approx \frac{1}{g_{m1}} \quad (22)$$

Accordingly, the differential transimpedance gain in the IF band is

$$Z_{t,d}(s) \approx \frac{2R_{eq}}{1 + sR_{eq}C_{eq}}, \quad Z_{t,d} \approx 2R_{eq} \quad (23)$$

with the dominant pole given by

$$f_p \approx \frac{1}{2\pi R_{eq}C_{eq}} \quad (24)$$

Here, R_L is the load resistance, R_{eq} is the equivalent output resistance, C_{eq} is the equivalent output capacitance, including the parasitic, load, and cross-coupled capacitances, and g_{m1} and g_{mb1} denote the transconductance and body-effect transconductance of the input transistor, respectively.

The dominant noise contributors are the load-resistor thermal noise and the channel thermal noise of the input devices. The corresponding input-referred current-noise spectral density can be approximated as

$$S_{i,in,ref} \approx \frac{2kT}{R_L} + 2kT\gamma(g_{m1} + g_{m2}) \quad (25)$$

where k is the Boltzmann constant, T is the absolute temperature, γ is the channel thermal-noise coefficient, and g_{m2} is the transconductance of the bias transistor.

Equations (23)–(25) indicate that increasing R_L improves the transimpedance gain and suppresses the resistor-related input-referred noise, but at the expense of bandwidth. Likewise, lowering the transistor noise generally requires a larger transconductance and hence higher power consumption. Therefore, the proposed TIA design is governed by a tradeoff among gain, bandwidth, noise, and power.

Figure 10 shows the small-signal magnitude response of the proposed common-gate current-mode TIA under different load resistances and bias currents at the TT corner and 27 °C in the HLMC 55 nm 1P4M CMOS process. In Figure 10a, the load resistance is varied from 2 kΩ to 16 kΩ, while the bias current is fixed at 20 μA. The corresponding gain at 1 MHz changes from −4.54 dB to 3.5 dB. In Figure 10b, the bias current is varied from 2 μA to 16 μA with the load resistance fixed at 13.9 kΩ, and the gain at 1 MHz increases from −7.2 dB to 4.52 dB. These results confirm that the proposed TIA provides both effective gain tuning and a clear low-pass response.

The observed behavior is consistent with the differential transimpedance relation

$$Z_{t,d}(s) \approx \frac{2R_{eq}}{1 + sR_{eq}C_{eq}}, \quad Z_{t,d} \approx 2R_{eq} \approx 2R_L \quad (26)$$

which indicates that the gain is mainly determined by the effective load resistance, while the pole frequency is jointly set by R_{eq} and C_{eq} . Therefore, the proposed common-gate TIA not only provides transimpedance amplification for the mixer output current, but also suppresses higher-order harmonics through its low-pass characteristic. Moreover, its moderate gain is beneficial for cascading with the following image-rejection filter, avoiding excessive front-end amplification that would otherwise limit the output swing and degrade linearity. Combined with the low-current bias strategy, the proposed TIA is suitable for the target low-power BLE receiver front-end.

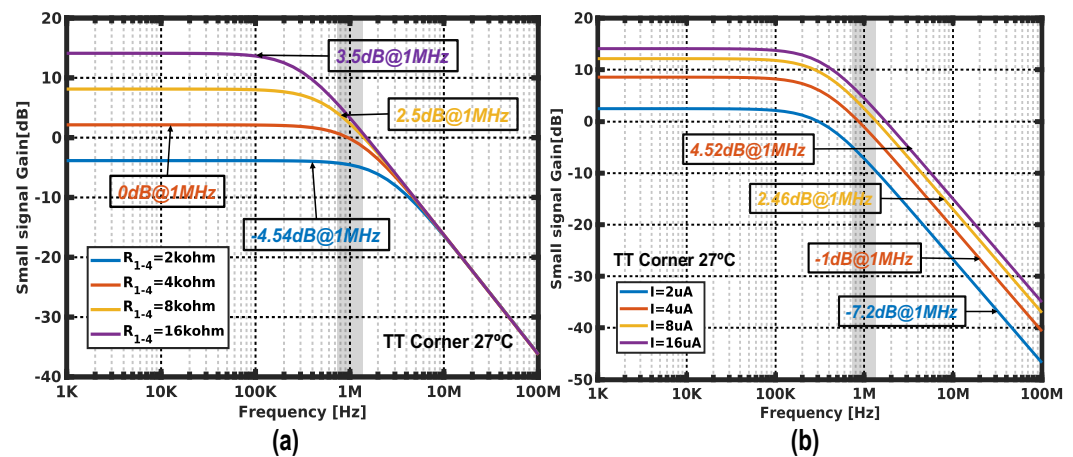


Figure 10. Small-signal magnitude response of the proposed common-gate current-mode TIA under different load resistances and bias currents: (a) for different load resistances and (b) for different bias currents.

3.3. Gm-C Complex Filter

In a Low-IF receiver, the desired signal and its image are translated to nearby frequencies after down-conversion, and a conventional low-pass response cannot effectively suppress the image component. Therefore, an image-rejection stage is required after the TIA. In this work, a Gm-C-based complex filter is adopted because it provides low power consumption, tunable center frequency, and effective image suppression. As shown in Figure 11, the quadrature-coupling transconductance shifts the real low-pass response to a complex bandpass response. To further enhance the image rejection and compensate for the moderate gain of the preceding TIA, a positive-feedback Gm-C complex filter is employed, as shown in Figure 11. The effective damping term and frequency shift are given by

$$\omega_{RE} = \frac{g_{m,RE} - g_{m,NEG}}{C}, \quad \omega_{shift} = \frac{g_{m,IM}}{C} \quad (27)$$

where $g_{m,RE}$ is the real-part transconductance, $g_{m,NEG}$ is the equivalent positive-feedback transconductance, $g_{m,IM}$ is the quadrature-coupling transconductance, and C is the integrating capacitance. The positive- and negative-sequence transfer functions can be written as

$$H_+(s) = \frac{g_m/C}{s + \omega_{RE} - j\omega_{shift}}, \quad H_-(s) = \frac{g_m/C}{s + \omega_{RE} + j\omega_{shift}} \quad (28)$$

where g_m is the input transconductance of the filter cell. Accordingly, the image rejection ratio is

$$IRR = \sqrt{1 + \left(\frac{2\omega_{shift}}{\omega_{RE}}\right)^2} = \sqrt{1 + 4Q^2}, \quad Q = \frac{g_{m,IM}}{g_{m,RE} - g_{m,NEG}} = \frac{\omega_{shift}}{\omega_{RE}} \quad (29)$$

Equation (29) indicates that the image-rejection performance can be improved either by increasing $g_{m,IM}$ or by reducing the effective damping term through positive feedback. This mechanism forms the basis of the proposed low-power image-rejection filter.

Figure 12 shows the simulated magnitude response of the cascaded two-stage complex filter in Figure 11 under the TT corner at 27 °C in the HLMC 55 nm CMOS process. The proposed filter provides a gain of 50 dB at 1 MHz, while the image gain at −1 MHz is 18.5 dB, corresponding to an IRR of 31.5 dB. These results verify that the proposed positive-feedback Gm-C complex filter achieves effective image suppression while maintaining sufficient passband gain, making it suitable for the proposed Low-IF receiver front-end.

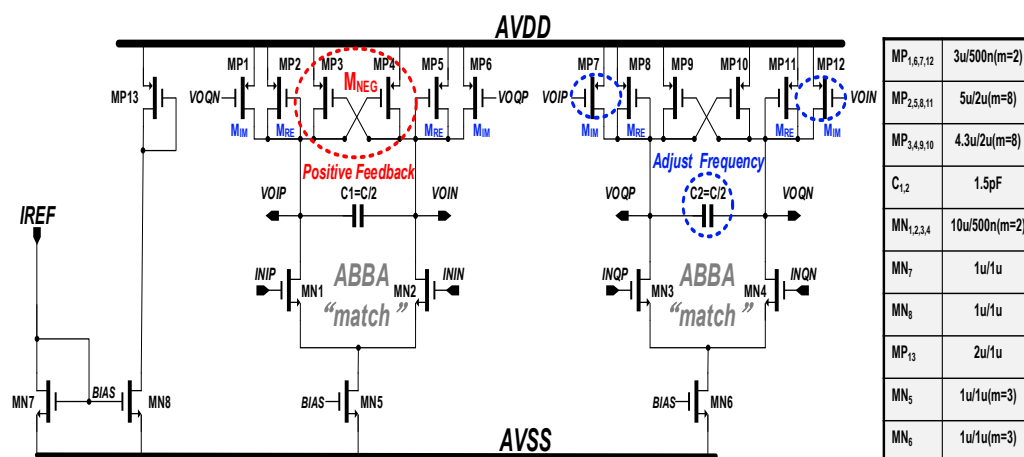


Figure 11. Complex filter implemented using a positive-feedback Gm-C cell for Low-IF image suppression.

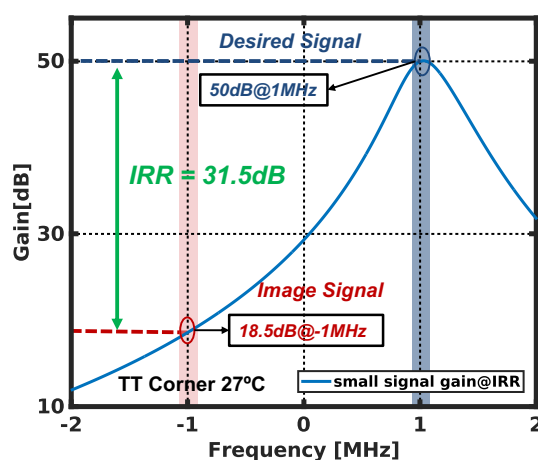


Figure 12. Simulated magnitude response of the Gm-C complex filter, illustrating passband shaping for the desired IF signal and suppression of the image-frequency component.

4. Results

4.1. Layout of the Complete Receiver Front-End

The layout of the proposed BLE receiver front-end is shown in Figure 13. Implemented in the HLMC 55 nm 1P4M CMOS process, the circuit occupies an active core area of $458 \mu\text{m} \times 177 \mu\text{m}$, corresponding to 0.081 mm^2 excluding bonding pads. As illustrated in the floorplan, the layout consists of four main blocks: the RF front-end integrating the LNTA and passive mixer, the common-gate TIA, the complex filter, and the MOS-capacitor block for supply stabilization. The RF front-end is placed on the left side to reduce high-frequency routing length, while the TIA occupies the central region as the main signal-conditioning stage between down-conversion and image-rejection filtering. The complex filter is located close to the TIA output to shorten the low-IF interconnect and alleviate parasitic degradation. This floorplan supports compact routing and is consistent with the low-power, small-area design target of the proposed receiver front-end.

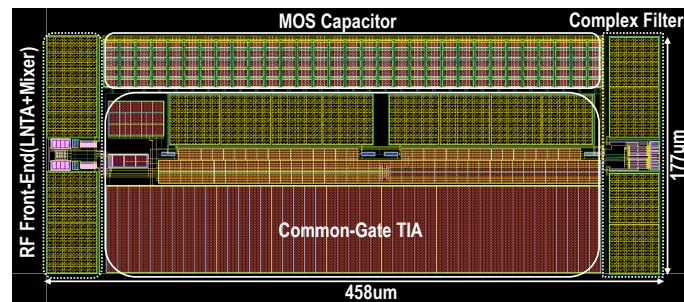


Figure 13. Layout of the proposed BLE receiver front-end implemented in a 55 nm CMOS process.

4.2. Post-Layout Results of the Complete Receiver Front-End

After parasitic extraction, the complete receiver front-end was verified by post-layout simulations under multiple process corners, including TT, FF, SS, and SF, and over a temperature range from -40°C to 80°C . The simulations were conducted to evaluate the robustness of the proposed design in terms of input matching, gain, noise figure, and image rejection ratio under process and temperature variations. In addition, the performance of the proposed receiver front-end was compared with previously reported domestic and international works in order to highlight its overall design tradeoff and practical advantages for low-power BLE applications.

4.2.1. Input Matching

Figure 14 presents the post-layout simulated S_{11} of the proposed receiver front-end with external impedance matching realized by a 4.7 nH inductor and a 20 pF capacitor. Simulations were carried out under the TT, FF, SS, and SF process corners at -40°C , 27°C , and 80°C . The gray-shaded region indicates the BLE operating band. In all cases, the input reflection coefficient remains below -10 dB over the BLE band, indicating satisfactory input matching. The TT and SF corners show better matching behavior, whereas the FF and SS corners exhibit noticeable center-frequency shifts with temperature due to variations in the input impedance. In future silicon measurement, the matching performance may be further improved by fine-tuning the matching network and considering the parasitic effects of the PCB, chip pads, and bonding wires.

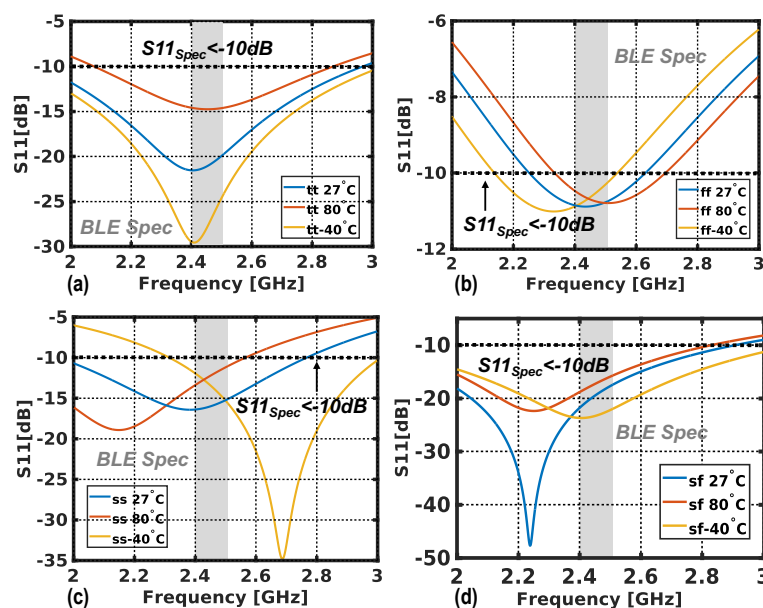


Figure 14. Post-layout simulated S_{11} of the proposed receiver front-end under different process corners: (a) TT, (b) FF, (c) SS, and (d) SF.

4.2.2. Noise Figure

Figure 15 shows the post-layout double-sideband noise figure of the proposed receiver front-end obtained by harmonic-balance noise analysis at an RF of 2.451 GHz, an input power of -60 dBm, and an LO frequency of 2.45 GHz with a 300 mV amplitude. Simulations were performed from 1 kHz to 10 MHz IF bandwidth under the TT, FF, SS, and SF process corners at -40 °C, 27 °C, and 80 °C. At an IF of 1 MHz, all cases satisfy the design target of $NF < 15$ dB. The minimum noise figure is achieved at the FF corner with 6.7 dB, while the worst case occurs at the SS corner with 14.2 dB. The variation in noise performance is attributed to process- and temperature-dependent changes in device parameters, passive components, and input impedance. Nevertheless, the proposed front-end maintains acceptable noise performance across all investigated PVT conditions.

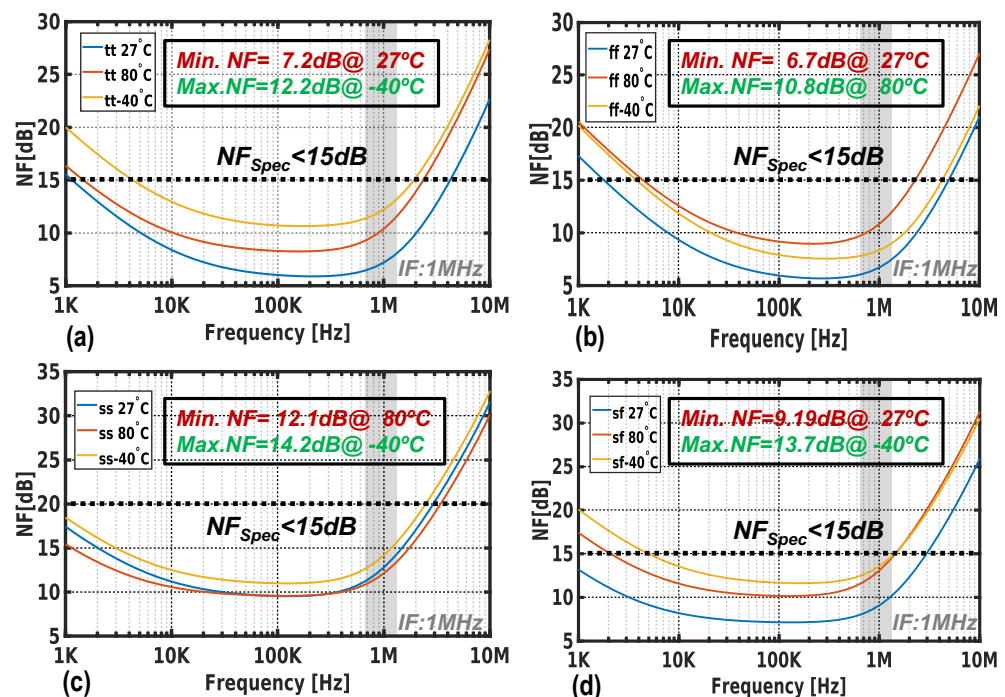


Figure 15. Post-layout double-sideband noise figure of the proposed receiver front-end under different process corners: (a) TT, (b) FF, (c) SS, and (d) SF.

4.2.3. Gain

Figure 16 presents the post-layout gain tuning behavior of the proposed receiver front-end at the TT corner and 27 °C. The simulations were performed at an RF of 2.451 GHz, an input power of -60 dBm, and an LO frequency of 2.45 GHz with a 300 mV amplitude. By varying the bias current of the baseband amplifier, the gain at an IF of 1 MHz can be adjusted from 18.7 dB at 1.25 μ A to 45.2 dB at 20 μ A. This result confirms the effectiveness of the proposed variable-gain scheme and indicates that the receiver input-power range can be extended by adaptive gain control.

Figure 17 presents the post-layout down-conversion voltage gain of the proposed receiver front-end obtained by harmonic-balance AC analysis at an RF of 2.451 GHz, an input power of -60 dBm, and an LO frequency of 2.45 GHz with a 300 mV amplitude. Simulations were performed under the TT, FF, SS, and SF process corners at -40 °C, 27 °C, and 80 °C. The FF corner achieves the highest gain of 46.4 dB at an IF of 1 MHz, while the TT corner provides 45.2 dB at the same IF. The SS corner shows the most degraded gain response. Frequency shifts are observed under the TT corner at -40 °C, the SS corner, and the SF corner at -40 °C, mainly due to variations in the input impedance and the center

frequency of the image-rejection filter. These offsets may be compensated in future designs by using trimming capacitance in the filter and by refining the input matching network.

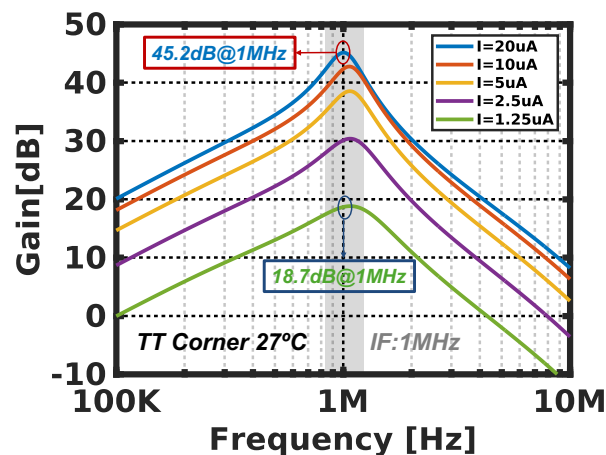


Figure 16. Post-layout gain tuning characteristic of the proposed receiver front-end at the TT corner and 27 °C under different baseband-amplifier bias currents.

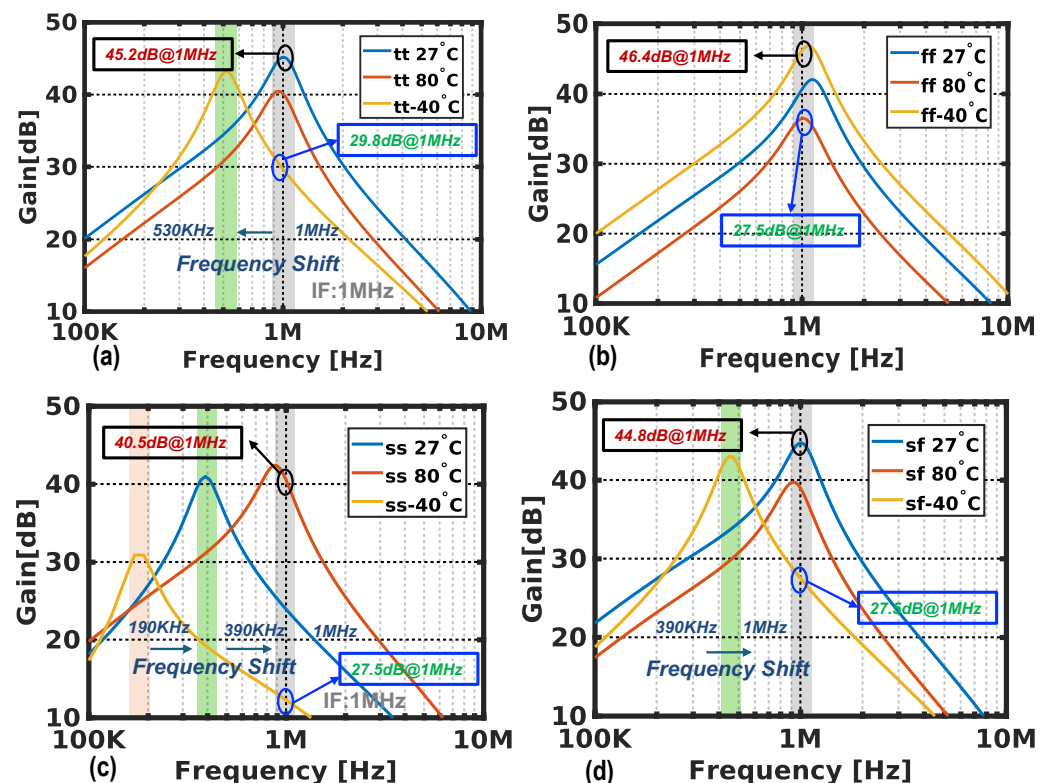


Figure 17. Post-layout down-conversion voltage gain of the proposed receiver front-end under different process corners: (a) TT, (b) FF, (c) SS, and (d) SF.

4.2.4. Linearity

The in-band linearity of the proposed receiver front-end was evaluated by the input third-order intercept point (IIP3). Two equal-amplitude tones at 2.455 GHz and 2.458 GHz were applied at the input, while the LO frequency was set to 2.45 GHz with a 300 mV amplitude. Using harmonic-balance AC analysis, the input power was swept to extract the post-layout IIP3 under the TT, FF, SS, and SF process corners at -40 °C, 27 °C, and 80 °C. The results, summarized in Table 3, show that all cases satisfy the linearity

target of $IIP3 > -28$ dBm, confirming acceptable in-band linearity across process and temperature variations.

Table 3. Post-layout simulated in-band $IIP3$ of the proposed receiver front-end under different process corners and temperatures.

Temperature	TT (dBm)	FF (dBm)	SS (dBm)	SF (dBm)
−40 °C	−15.06	−17.60	−19.51	−15.63
27 °C	−16.95	−17.99	−12.61	−16.61
80 °C	−16.98	−18.07	−15.87	−16.26

4.2.5. Image rejection ratio

Figure 18 shows the post-layout image-rejection performance of the proposed receiver front-end at the TT corner and 27 °C. Under an RF input frequency of 2.451 GHz, an input power of −60 dBm, and an LO frequency of 2.45 GHz with a 300 mV amplitude, the desired signal gain is 45.2 dB at 2.451 GHz, while the image signal gain is 17.1 dB at 2.499 GHz. The resulting image rejection ratio is 28.1 dB, which is higher than the design target of 24 dB. This result verifies the effectiveness of the proposed complex-filter-based image-rejection scheme in the Low-IF receiver front-end.

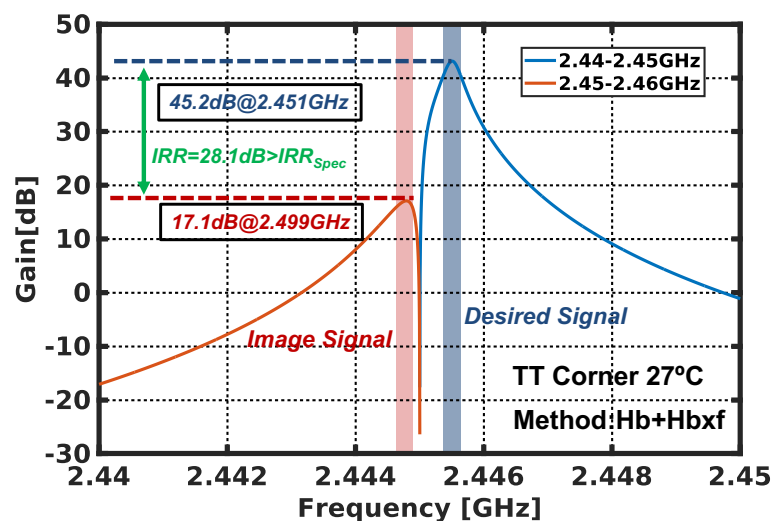


Figure 18. Post-layout image-rejection performance of the proposed receiver front-end at the TT corner and 27 °C.

4.2.6. Power-Consumption Breakdown

Figure 19 shows the post-layout power breakdown of the proposed receiver front-end at the TT corner and 27 °C. As indicated in the figure, the LNTA and passive mixer together consume 200.8 μ W, accounting for approximately 37% of the total power. The common-gate TIA consumes 268.8 μ W, corresponding to 50% of the total power, while the image-rejection complex filter (IRA) consumes 67.4 μ W, accounting for the remaining 13%. The total power consumption of the complete receiver front-end is 537 μ W. These results show that the TIA is the dominant power-consuming block in the proposed architecture, whereas the image-rejection stage contributes only a modest portion of the overall power budget.

Receiver Power Distribution (Total Power Consumption:537uW)

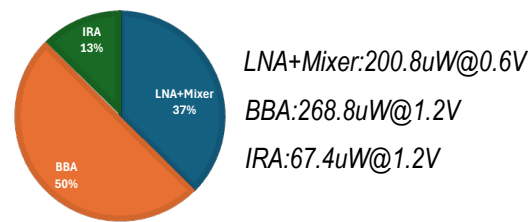


Figure 19. Post-layout power breakdown of the proposed receiver front-end at the TT corner and 27 °C.

5. Discussion

The proposed receiver front-end is intended for low-power BLE body area network (BAN) sensor nodes, where short-range communication, strict energy constraints, and compact implementation are key requirements. In this work, a 1 MHz Low-IF architecture is adopted as a practical compromise between circuit simplicity and robustness. The main contribution is not the Low-IF architecture itself, but the BAN-oriented co-design of the RF front-end, common-gate TIA, and Gm-C complex filter under low-power and small-area constraints. In particular, the self-biased inverter-based LNTA with native MOS devices supports low-voltage operation, while the removal of the conventional interstage coupling capacitor reduces area and parasitic loading.

The post-layout results show that the proposed front-end achieves 45.2 dB gain, 7.2 dB NF, and 28.1 dB IRR over the 2.4–2.48 GHz band with a total power consumption of 537 μ W, meeting the adopted BAN-oriented design targets. As summarized in Table 4, the proposed design achieves a favorable overall tradeoff among gain, NF, IRR, power consumption, and area. In particular, the FOM reaches 48.1, which is the highest among the compared works [7,12,15,30,31], confirming an advantageous gain–noise–power tradeoff. Although the in-band linearity is still inferior to that of some prior designs [12,30,31], the overall results indicate that the proposed front-end is attractive for low-power BLE BAN applications.

Table 4. Performance summary and comparison of the BLE receiver front-end.

Parameter	TICASII-21 [15]	TMTT-2 [7]	JSSC-23 [12]	MWCL-24 [30]	ISCAS-25 [31]	This Work (Post Sim.)
Frequency (GHz)	2.4–2.48	2.4–2.48	2.4–2.48	2.4–2.48	2.4	2.4–2.48
Architecture	Low-IF	Low-IF	Zero-IF	Uncertain-IF	High-IF	Low-IF
Baseband Topology	With IQ	With IQ	With IQ	Without IQ	With IQ	With IQ
NF (dB)	8.16	8.2	4.5	12.9	15	7.2
IB-IIP3 (dBm)	−27.5	−25.75	−12.0 ^(b)	−6.2 ^(b)	−4.3 ^(b)	−12.61 ^(b)
Power (mW) ^(c)	1.52	2.16	0.61	0.129	0.243	0.537
Gain (dB)	54.5	49.5	39.75	40.7	26	45.2
Supply (V)	0.8	0.8	0.53	0.8	0.8	0.6/1.2
Area (mm ²)	1.65	1.16	0.8	0.456	0.415	0.081
Process (CMOS)	65 nm	65 nm	28 nm	55 nm	28 nm	55 nm
FOM ^(a)	47.27	43.21	47.5 ^(b)	46.34	34.1 ^(b)	48.1

^(a) $FOM = 10 \log_{10} \left(\frac{10^{Gain/20}}{Power \cdot 10^{NF/10}} \right)$; ^(b) RF front-end result; ^(c) excluding the LO buffer power consumption.

Some limitations should also be noted. The present work is based on post-layout verification, while silicon measurement results are not yet available. In addition, mismatch- and yield-related effects still require further investigation through future silicon validation. Future work will focus on silicon measurement and further improvement of linearity through optimization of the LNTA, TIA, and image-rejection circuitry.

6. Conclusions

A low-power Low-IF BLE receiver front-end for BAN applications has been presented. The proposed design adopts a self-biased inverter-based LNTA, a single-balanced passive mixer, a common-gate TIA, and a Gm-C complex filter to achieve a compact and low-power receiver architecture. By using native MOS devices and removing the conventional interstage coupling capacitor, the front-end supports low-voltage operation with reduced area and parasitic loading. Implemented in a 55 nm standard CMOS process, the proposed front-end occupies 0.081 mm² active area excluding bonding pads and achieves 45.2 dB gain, 7.2 dB NF, and 28.1 dB IRR over the 2.4–2.48 GHz band with 537 μ W power consumption in post-layout simulations. These results verify that the proposed design is suitable for low-power BLE BAN sensor nodes.

Author Contributions: Conceptualization, Y.X. and Z.Z.; circuit design, Y.X.; validation, Y.X. and L.L.; writing—original draft preparation, Y.X.; writing—review and editing, L.L. and Z.Z.; supervision, L.L. All authors have read and agreed to the published version of the manuscript.

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Abbreviations

The following abbreviations are used in this manuscript:

RF	Radio frequency
IF	Intermediate frequency
Low-IF	Low-intermediate-frequency
Zero-IF	Zero-intermediate-frequency
DDS	Direct Digital Synthesizer
PLL	Phase-Locked Loop
LO	Local Oscillator
IIP3	Input-referred third-order intercept point
IRR	Image rejection ratio
NF	Noise figure
CMOS	Complementary metal–oxide–semiconductor
PVT	Process, voltage, and temperature
PCB	Printed circuit board
PAD	Pad
EEG	Electroencephalography
ECG	Electrocardiography
EMG	Electromyography
BP	Blood pressure
SpO ₂	Peripheral capillary oxygen saturation

IMU	Inertial measurement unit
DC	Direct current

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